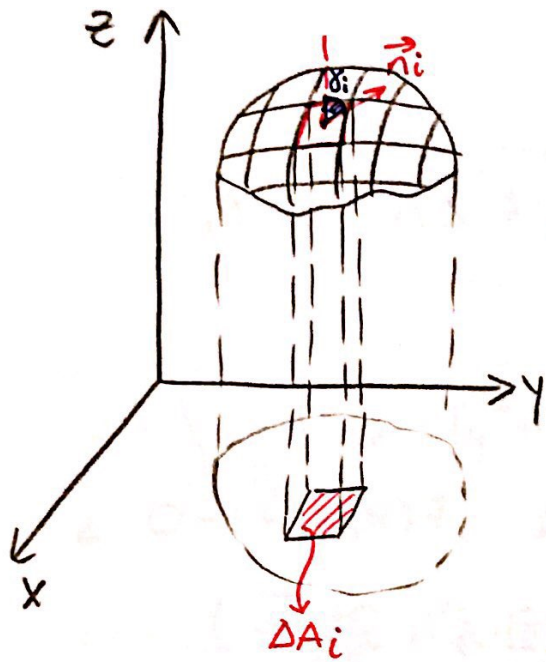


CALCULATION OF SURFACE AREAS



let the smooth surface S be given by $z = f(x, y)$. Consider the partition of the surface by the parallel planes to the coordinate axes.

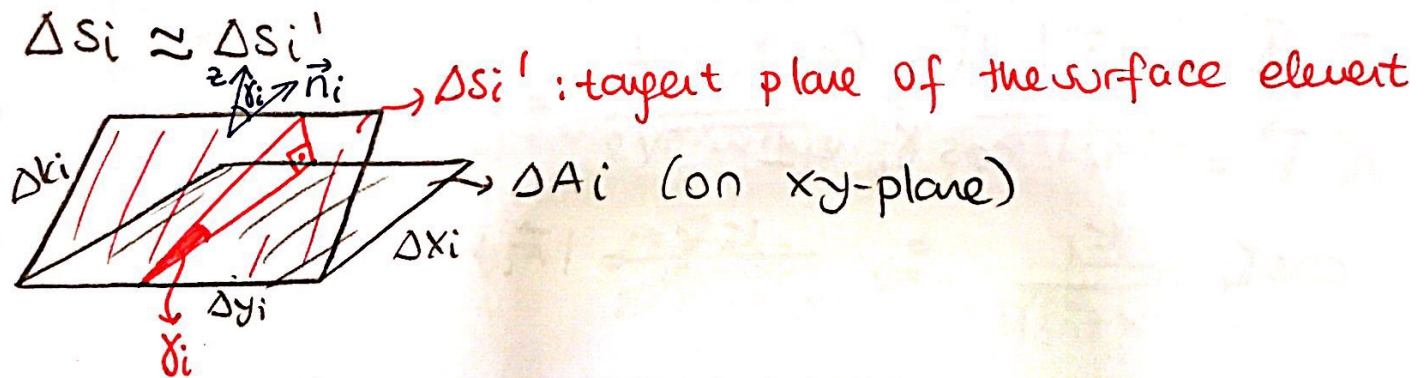
$$\Delta S_i \quad (i=1, 2, \dots, n)$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta S_i = S.$$

let the projection of surface element ΔS_i be ΔA_i onto xy -plane. According to the type of partition of S , these projections are rectangular regions.

$$\Delta A_i = \Delta x_i \Delta y_i.$$

Now consider the tangent planes $\Delta S_i'$ instead of the surface elements ΔS_i . (approximation)



$$S = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta S_i'$$

let denote the normal of the tangent planes with $\vec{n}_i$.

$$\left. \begin{aligned} \Delta A_i &= \Delta x_i \cdot \Delta y_i \\ \Delta S_i' &= \Delta k_i \cdot \Delta y_i \end{aligned} \right\} \cos \gamma_i = \frac{\Delta x_i}{\Delta k_i} \Rightarrow \Delta k_i = \Delta x_i \cdot \sec \gamma_i$$

$$\Delta S_i' = \Delta k_i \cdot \Delta y_i = \Delta x_i \cdot \Delta y_i \cdot \sec \gamma_i = \Delta A_i \cdot \sec \gamma_i$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \sec \gamma_i \cdot \Delta A_i$$

Now we have to find γ_i :

Normal vector of the surface $F(x, y, z) = 0$ is

$$\vec{n}_i = \text{grad } F = \nabla F \quad \left(\nabla F = \frac{\partial F}{\partial x} \vec{i} + \frac{\partial F}{\partial y} \vec{j} + \frac{\partial F}{\partial z} \vec{k} \right)$$

If the equation is given by $z = f(x, y)$ then

$$F = z - f(x, y) = 0$$

$\downarrow$
 $F(x, y, z) = z - f(x, y) = 0$ and the normal

$$\vec{n}_i = \nabla F = -\frac{\partial f}{\partial x} \vec{i} - \frac{\partial f}{\partial y} \vec{j} + \vec{k}$$

$$\vec{n}_i \cdot \vec{k} = |\vec{n}_i| \cdot |\vec{k}| \cdot \cos \gamma_i = 1$$

$$\vec{n}_i \cdot \vec{k} = |\vec{n}_i| \cdot \cos \gamma_i = 1$$

$$\cos \gamma_i = \frac{1}{|\vec{n}_i|} \Rightarrow \sec \gamma_i = |\vec{n}_i|$$

$$\sec \gamma_i = |\vec{n}_i| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1}$$

$$S = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{f_x^2 + f_y^2 + 1} \cdot \Delta A_i$$

$$S = \iint_D \sqrt{f_x^2 + f_y^2 + 1} \, dx \, dy$$

Surface area when the projection is on xy -plane.

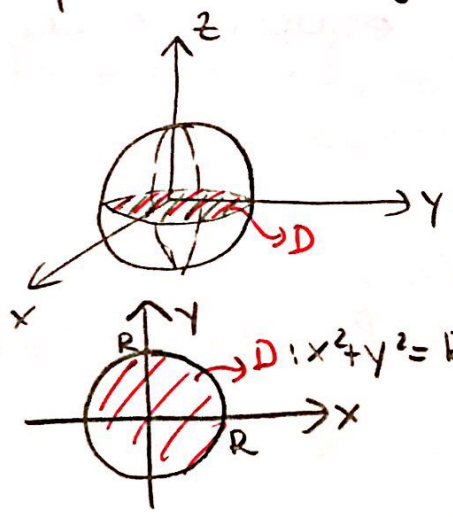
$$S = \iint_{D_1} \sqrt{f_x^2 + f_z^2 + 1} \, dx \, dz$$

when the projection is on xz -plane. ($\sec \alpha$)

$$S = \iint_{D_2} \sqrt{f_z^2 + f_y^2 + 1} \, dy \, dz$$

when the projection is on yz -plane. ($\sec \beta$)

Example: Evaluate the surface area of the sphere $x^2 + y^2 + z^2 = R^2$.



$$x^2 + y^2 + z^2 = R^2$$

$$z = \pm \sqrt{R^2 - x^2 - y^2}$$

$$\frac{S}{2} = \iint_D \sqrt{f_x^2 + f_y^2 + 1} \, dx \, dy$$

$$z = f(x, y) = \sqrt{R^2 - x^2 - y^2}$$

$$f_x = \frac{-2x}{2\sqrt{R^2 - x^2 - y^2}} = \frac{-x}{\sqrt{R^2 - x^2 - y^2}}$$

$$f_y = \frac{-2y}{2\sqrt{R^2 - x^2 - y^2}} = \frac{-y}{\sqrt{R^2 - x^2 - y^2}}$$

$$\frac{S}{2} = \iint_D \sqrt{\frac{x^2 + y^2}{R^2 - x^2 - y^2} + 1} \, dx \, dy, \text{ use polar transf.}$$

$$\frac{S}{2} = \int_0^{2\pi} \int_0^R \sqrt{\frac{r^2}{R^2 - r^2} + 1} \, r \, dr \, d\theta, \quad \begin{matrix} x = r \cos \theta \\ y = r \sin \theta \end{matrix}$$

$$\frac{S}{2} = \int_0^{2\pi} \int_0^R \sqrt{\cancel{r^2} + R^2 - \cancel{r^2}} r dr d\theta = \int_0^{2\pi} \int_0^R \frac{R}{\sqrt{R^2 - r^2}} r dr d\theta$$

$$R^2 - r^2 = t^2$$

$$-\cancel{r} dr = \cancel{2} t dt \Rightarrow r dr = -t dt$$

$$\frac{S}{2} = \int_0^{2\pi} \left(\int_R^0 \frac{R(-\cancel{t} dt)}{\cancel{t}} \right) d\theta = R \int_0^{2\pi} \left(\int_0^R dt \right) d\theta$$

$$\frac{S}{2} = R \cdot \int_0^{2\pi} R d\theta \Rightarrow \frac{S}{2} = R^2 \cdot 2\pi \Rightarrow \underline{S = 4\pi R^2}$$

If Surface is given by Parametric Equations:

If S is defined by parametric equations as;

$$x = f(u, v)$$

$$y = g(u, v)$$

$$z = h(u, v)$$

In this case the surface area of $\mathcal{C}(x, y, z) = 0$ is;

$$S = \iint_D \sqrt{EG - F^2} du dv$$

$$* E = \left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial u} \right)^2$$

$$* G = \left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2$$

$$* F = \frac{\partial x}{\partial u} \cdot \frac{\partial x}{\partial v} + \frac{\partial y}{\partial u} \cdot \frac{\partial y}{\partial v} + \frac{\partial z}{\partial u} \cdot \frac{\partial z}{\partial v}$$

Example: If the sphere $x^2 + y^2 + z^2 = R^2$ is given by parametric equations as;

$$\begin{cases} x = R \sin \phi \cos \theta \\ y = R \sin \phi \sin \theta \\ z = R \cos \phi \end{cases}$$

then evaluate the surface area of it.

$$S = \iint_D \sqrt{EG - F^2} \, d\phi \, d\theta$$

$$* E = (R \cos \phi \cos \theta)^2 + (R \cos \phi \sin \theta)^2 + (-R \sin \phi)^2$$

$$E = R^2 \cos^2 \phi \cos^2 \theta + R^2 \cos^2 \phi \sin^2 \theta + R^2 \sin^2 \phi$$

$$E = R^2 \cos^2 \phi (\underbrace{\cos^2 \theta + \sin^2 \theta}_1) + R^2 \sin^2 \phi = R^2$$

$$* G = (-R \sin \phi \sin \theta)^2 + (R \sin \phi \cos \theta)^2 + 0^2$$

$$G = R^2 \sin^2 \phi \sin^2 \theta + R^2 \sin^2 \phi \cos^2 \theta = R^2 \sin^2 \phi$$

$$* F = (R \cos \phi \cos \theta)(-R \sin \phi \sin \theta) + (R \cos \phi \sin \theta)(R \sin \phi \cos \theta) + 0 \cdot (-R \sin \phi)$$

$$F = -R^2 \sin \phi \cos \phi \cos \theta \sin \theta + R^2 \cos \phi \sin \phi \sin \theta \cos \theta = 0$$

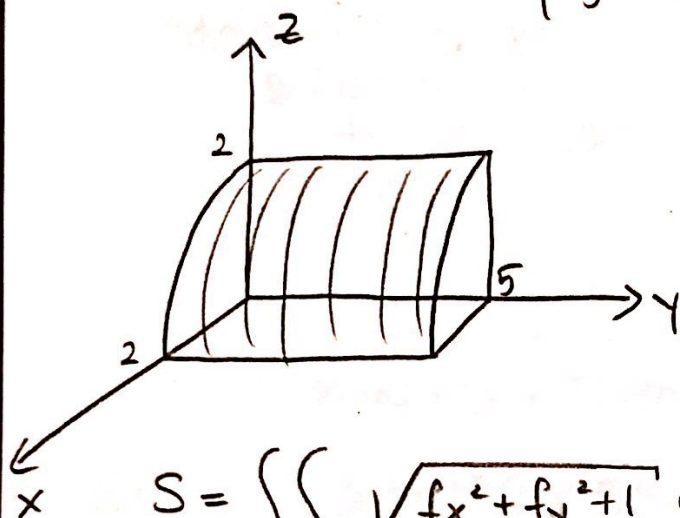
$$S = \iint_D \sqrt{R^2 \cdot R^2 \sin^2 \phi - 0} \, d\phi \, d\theta = \iint_D R^2 \sin \phi \, d\phi \, d\theta$$

$$S = \int_0^{2\pi} \int_0^\pi R^2 \sin \phi \, d\phi \, d\theta = -R^2 \int_0^{2\pi} \cos \phi \Big|_0^\pi \, d\theta$$

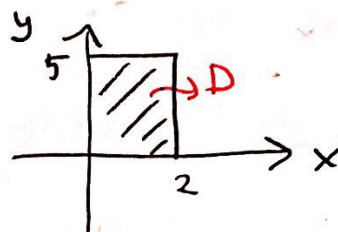
$$S = -R^2 \int_0^{2\pi} (\underbrace{\cos \pi}_{-1} - \underbrace{\cos 0}_1) \, d\theta = -R^2 \cdot (-2) \cdot 2\pi$$

$$\underline{S = 4\pi R^2}$$

Example: Evaluate the area of the surface that is enclosed by: $\begin{cases} x^2 + z^2 = 4 \\ y=0, x=0, z=0, y=5 \end{cases}$.



Projection onto xy-plane



$$z = \pm \sqrt{4-x^2} = f(x, y)$$

$$S = \iint_D \sqrt{f_x^2 + f_y^2 + 1} \, dx \, dy, \quad z_x = \frac{-x}{\sqrt{4-x^2}}, \quad z_y = 0$$

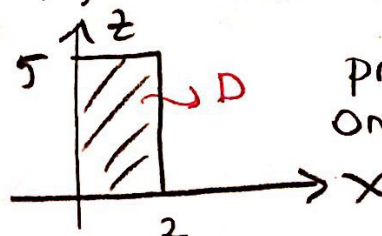
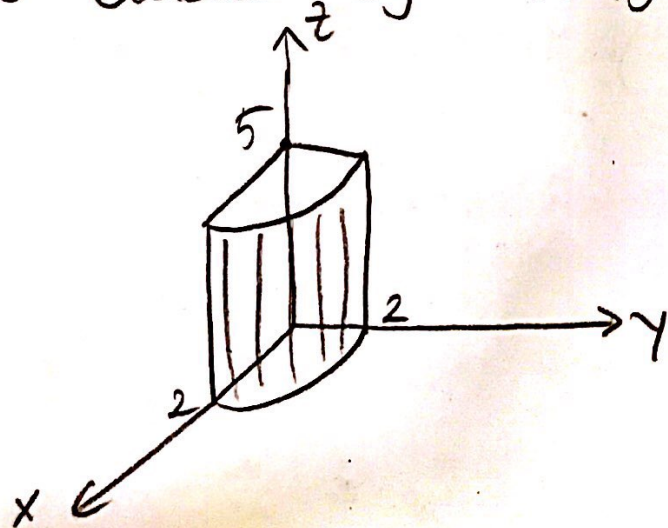
$$S = \iint_D \sqrt{\frac{x^2}{4-x^2} + 0 + 1} \, dx \, dy = \iint_D \sqrt{\frac{x^2 + 4 - x^2}{4-x^2}} \, dx \, dy$$

$$S = \iint_D \frac{2}{\sqrt{4-x^2}} \, dx \, dy = \int_0^5 \int_0^2 \frac{2}{\sqrt{4-x^2}} \, dx \, dy = 2 \int_0^5 \arcsin \frac{x}{2} \Big|_0^2 \, dy$$

$$S = 2 \int_0^5 \left(\frac{\pi}{2} - 0 \right) dy = 2 \cdot \frac{\pi}{2} \cdot 5$$

$$\underline{S = 5\pi}$$

Example: Evaluate the area of the surface that is enclosed by $x^2 + y^2 = 4, x=0, y=0, z=0, z=5$.



projection onto xz-plane.

$$x^2 + y^2 = 4 \Rightarrow y = \pm \sqrt{4-x^2}$$

$$S = \iint_D \sqrt{f_x^2 + f_z^2 + 1} \, dx \, dz, \quad f_x = \frac{-x}{\sqrt{4-x^2}}, \quad f_z = 0$$

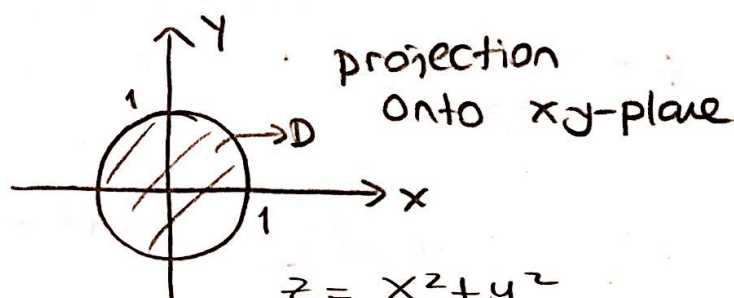
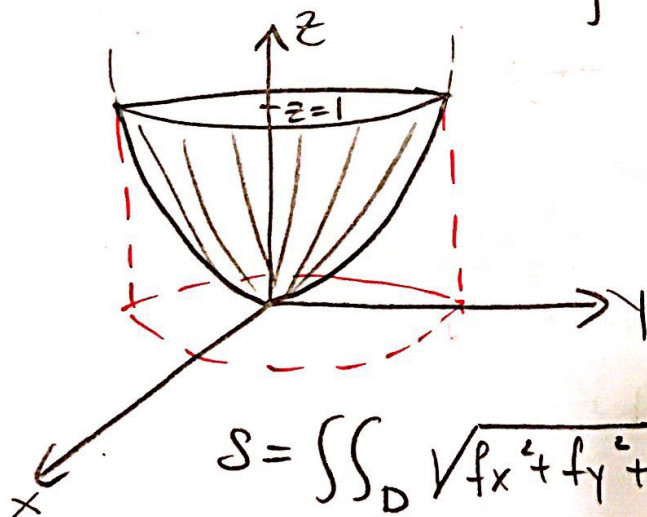
$$S = \iint_D \sqrt{\frac{x^2}{4-x^2} + 0 + 1} \, dx \, dz = \iint_D \sqrt{\frac{x^2 + 4 - x^2}{4-x^2}} \, dx \, dz$$

$$S = \iint_D \frac{2}{\sqrt{4-x^2}} \, dz \, dx = \int_0^2 \int_0^5 \frac{2}{\sqrt{4-x^2}} \, dz \, dx$$

$$S = \int_0^2 \frac{2z}{\sqrt{4-x^2}} \Big|_0^5 \, dx = \int_0^2 \frac{10}{\sqrt{4-x^2}} \, dx = 10 \cdot \text{Arcsin} \frac{x}{2} \Big|_0^2$$

$$S = 10 \cdot \left(\frac{\pi}{2} - 0 \right) \Rightarrow \underline{S = 5\pi}$$

Example: Evaluate the area of the surface that is enclosed by $z = x^2 + y^2$ and $z = 1$.



$$z = x^2 + y^2$$

$$\downarrow$$

$$f(x, y) = x^2 + y^2$$

$$f_x = 2x, \quad f_y = 2y$$

$$S = \iint_D \sqrt{f_x^2 + f_y^2 + 1} \, dx \, dy$$

$$S = \iint_D \sqrt{4x^2 + 4y^2 + 1} \, dx \, dy, \quad \text{with polar transformation } x = r \cos \theta, \quad y = r \sin \theta$$

$$S = \int_0^{2\pi} \int_0^1 \sqrt{4r^2 + 1} \, r \, dr \, d\theta$$

$$4r^2 + 1 = t^2$$

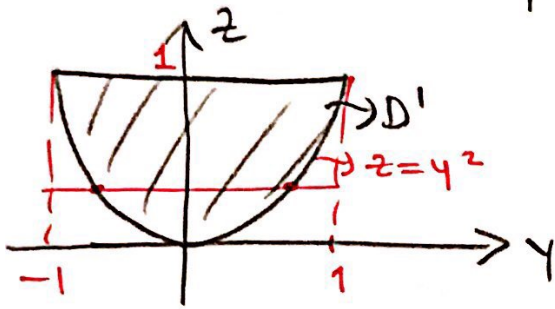
$$8r \, dr = 2t \, dt$$

$$r \, dr = \frac{1}{4} t \, dt$$

$$S = \frac{1}{4} \int_0^{2\pi} \left(\int_1^{\sqrt{5}} t \cdot t \, dt \right) d\theta = \frac{1}{4} \int_0^{2\pi} \frac{t^3}{3} \Big|_1^{\sqrt{5}} d\theta = \frac{1}{12} \int_0^{2\pi} (5\sqrt{5} - 1) d\theta$$

$$S = \left(\frac{5\sqrt{5} - 1}{12} \right) 2\pi \Rightarrow \underline{S = \frac{\pi}{6} (5\sqrt{5} - 1)}$$

2nd way! have projection on to yz -plane



$$z = x^2 + y^2$$

$$x^2 = z - y^2$$

$$x = \pm \sqrt{z - y^2}$$

$$f_x = \pm \sqrt{z - y^2}, \quad f_y = \frac{-y}{\sqrt{z - y^2}}$$

$$\frac{S}{2} = \iint_{D'} \sqrt{f_y^2 + f_z^2 + 1} \, dy \, dz$$

$$f_z = \frac{1}{2\sqrt{z - y^2}}$$

$$\frac{S}{2} = \iint_{D'} \sqrt{\frac{y^2}{z - y^2} + \frac{1}{4(z - y^2)} + 1} \, dy \, dz$$

$$\begin{aligned} \textcircled{*} \quad z &= y^2 \\ y &= \pm \sqrt{z} \end{aligned}$$

$$\frac{S}{2} = \iint_{D'} \sqrt{\frac{4y^2 + 1 + 4(z - y^2)}{4(z - y^2)}} \, dy \, dz$$

$$\frac{S}{2} = \frac{1}{2} \iint_{D'} \sqrt{\frac{4z^2 + 1 + 4z - 4y^2}{z - y^2}} \, dy \, dz$$

$$S = \iint_{D'} \sqrt{\frac{4z + 1}{z - y^2}} \, dy \, dz = \int_0^1 \int_{-\sqrt{z}}^{\sqrt{z}} \frac{\sqrt{4z + 1}}{\sqrt{z - y^2}} \, dy \, dz$$

$$S = \int_0^1 \sqrt{4z + 1} \cdot \left. \arcsin \frac{y}{\sqrt{z}} \right|_{-\sqrt{z}}^{\sqrt{z}} dz$$

$$S = \int_0^1 \sqrt{4z + 1} \cdot \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) dz = \pi \int_0^1 \sqrt{4z + 1} \, dz$$

$$S = \frac{\pi}{4} \int_1^5 t^{1/2} \cdot dt = \frac{\pi}{4} \cdot \frac{2}{3} \cdot t^{3/2} \Big|_1^5$$

$$\begin{aligned} 4z + 1 &= t \\ 4dz &= dt \end{aligned}$$

$$S = \frac{\pi}{6} \cdot (5\sqrt{5} - 1)$$